DEFT-DPT: A Lightweight Multimodal Framework for Egocentric Video Action Recognition

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***Abstract*—Egocentric video analysis has emerged as an active and rapidly growing area of research in computer vision and multimedia due to its ability to capture human actions from a first-person perspective rich in visual and motion cues. This paper presents a lightweight multimodal framework for action localization and recognition in egocentric videos. Existing approaches often fail to effectively leverage complementary information from RGB, motion, and depth modalities, and even when multiple cues are fused, they typically lack adaptive integration, accurate localization, and scalability for large datasets. To overcome these challenges, we introduce the Dynamic Egocentric Feature Transformation (DEFT) module, which adaptively encodes features while preserving the distinctive motion and viewpoint characteristics of egocentric videos. DEFT emphasizes on central and action-relevant regions through spatial alignment and Gaussian-based weighting, enabling precise action localization and robust recognition. Furthermore, a lightweight Co-Attention module adaptively fuses complementary modality features through a cross-modal agreement. Next, we construct a Sparse Video Similarity Graph (SVSG) in the deep feature space using ResNet-50, where the central frame regions serve as graph nodes. To maintain computational efficiency, Dynamic Percentile Thresholding (DPT) retains only the most informative graph edges, yielding a sparse, yet meaningful graph representation. The resulting graph is then processed by a Graph Attention Network (GAT) for context-aware learning and action classification. Our lightweight framework, containing only 34.21M parameters, achieves 46.25% lower computation time compared to large-scale pretrained foundation models, while maintaining competitive accuracy. Extensive experiments on four benchmark egocentric video datasets—EPIC-Kitchens (RGB + Optical Flow), MECCANO (RGB + Depth), GTEA (RGB), and ADL (RGB)—demonstrate the superiority of the proposed method, showing consistent improvements in different modalities and settings. Our implementation and code are publicly available at https://github.com/pawanesh-mnnit/deft_dpt**

***Index Terms*—Egocentric video, action recognition, dynamic egocentric feature transformation, sparse video similarity graph, graph attention network.**

I. Introduction

ACTION recognition has been a key research topic in computer vision and multimedia, particularly in the context of egocentric videos, focusing on detecting and classifying human activities [1], [2], [3]. Researchers have explored a wide range of approaches, from traditional hand-crafted descriptors [2], [3] to deep learning-based representations [4], [5] in egocentric videos. The hand-crafted feature based methods capture local visual patterns, whereas deep network learns hierarchical and discriminative spatio-temporal features from large-scale egocentric video datasets. Egocentric videos provide a natural first-person perspective [6] where important actions and objects typically appear near the frame center and are rarely occluded. With the camera mounted on the observer's head, handled objects appear close and large, which highlights the user's interactions with the environment. Egocentric videos are inherently more challenging due to dynamic camera motion, limited field of view, frequent occlusions, and continuous viewpoint shifts. Many recent approaches on egocentric action recognition [3], [7], [8] have made significant progress, but critical multimodal challenges still persists. Many of these methods are not fully end-to-end and lack explicit 3D reasoning, making them less reliable in the presence of ego-motion, occlusion, or complex hand–object interactions. They often rely on limited input modalities, such as RGB and optical flow, and use rigid fusion strategies that fail to capture deeper relationships across modalities. Moreover, these approaches frequently struggle to accurately localize regions of interest where hand–object interactions occur, resulting in weak attention to action specific areas and reduced interpretability. In addition, compositional models based on fixed structures, such as verb–noun–preposition, limit flexibility and adaptability across diverse scenarios.

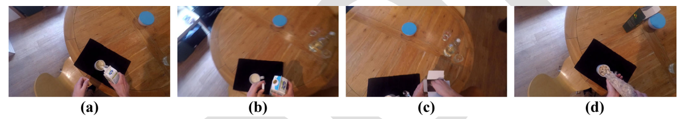

Fig. 1. Sample actions from the EPIC-kitchens dataset [9].

In Figure 1, we present sample actions from the EPIC-Kitchens dataset [9]. Figure 1(a-d), respectively illustrates, (a) "open milk:soy" (label 18), (b) "close milk:soy" (label 20), (c) "open bag:cereal" (label 27), and (d) "pour cereal" (label 30). These actions clearly shows that hand–object interactions predominantly occur near the frame center, reflecting a strong spatial bias in egocentric videos. However, existing methods [5], [6] often fail to fully leverage the complementary information available across RGB, motion, and depth modalities, thereby limiting their ability to model complex activities

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accurately. Although recent studies have explored multimodal approaches in egocentric videos [10], they generally struggle to integrate these cues and precisely localize actions adaptively. In real-world egocentric scenarios, spatial and motion information vary dynamically in response to hand movements, object interactions, and viewpoint changes, emphasizing the need for adaptive feature alignment and temporal reasoning across modalities rather than fixed integration. However, most existing studies have focused primarily on action recognition, with action localization remaining largely unexplored in the egocentric domain.

To address these limitations, in this paper, we first introduce a dynamic multimodal framework that jointly models spatial and temporal dependencies among different modalities, with a central focus on Dynamic Egocentric Feature Transformation (DEFT). Initially, the framework selects the available modalities RGB, motion (optical flow), and depth depending on each dataset, ensuring that complementary information is captured for subsequent processing. The framework then focuses on Dynamic Egocentric Feature Transformation (DEFT), which adaptively encodes features while preserving the distinctive motion and viewpoint characteristics of egocentric videos. Extracts features from central frame regions and generates weighted maps through spatial alignment and Gaussian-based weighting, enabling precise action localization and improved recognition accuracy. By focusing on the most informative regions—typically hands and manipulated objects, DEFT integrates a Spatial Transformer Network (STN) [4] for learnable affine transformations with Gaussian weighting, which encodes the egocentric spatial prior. This allow the adaptive spatial refinement while maintaining egocentric consistency. This module is optimized via a multi-task loss that aligns transformations, regularizes affine parameters, and enhances semantic weighting, resulting in accurate and robust action recognition.

Secondly, existing single- and multi-modal action recognition methods often lack scalability when applied to large egocentric video datasets. To overcome this, we develop a Sparse Video Similarity Graph (SVSG) in the deep feature space using ResNet-50, where frame regions serve as graph nodes. To retain the most informative connections while controlling graph density, we introduce Dynamic Percentile Thresholding (DPT), which enhances both scalability and computational efficiency. Finally, the resulting graph is processed by a Graph Attention Network (GAT) for training and classification, where action labels are predicted for each node. Unlike conventional Graph Convolutional Networks (GCNs), GAT employs an attention-based mechanism that adaptively weights neighboring nodes, effectively modeling the complex spatiotemporal dependencies. This adaptive learning enables the GAT to capture subtle temporal variations in egocentric sequences, leading to more accurate and computationally efficient action recognition. To further assess generalization and efficiency, we compare our DEFT–DPT framework with recent large-scale pretrained foundation models such as VideoMAE [11], EAKD [12], and EgoVLP [13]. Our model achieves competitive performance with significantly reduced computation time, demonstrating both efficiency and strong generalization capability for fine-grained egocentric action recognition.

The main contributions are now highlighted below:

1) **Dynamic Egocentric Feature Transformation (DEFT)**: We introduce DEFT, an adaptive feature encoding module that preserves the motion and viewpoint characteristics of egocentric videos. DEFT highlights informative regions, such as hands and manipulated objects, for accurate action localization and recognition. Furthermore, a lightweight Co-Attention module adaptively fuses complementary modality features based on cross-modal agreement. We also design a customized multi-task loss that jointly optimizes transformation alignment, affine regularization, and spatial weighting for stable and meaningful spatial attention learning.
2) **Sparse Video Similarity Graph (SVSG) with Dynamic Percentile Thresholding (DPT):** We develop a scalable graph-based representation that models frame-wise relationships in the deep feature space. DPT ensures that the graph retains the most informative edges while controlling density, improving computational efficiency for large egocentric datasets.
3) **Graph Attention Network (GAT) for Action Recognition:** The resulting SVSG is processed with GAT, which leverages attention-based learning to adaptively weight the neighboring nodes, which effectively captures the spatiotemporal dependencies. This leads to more accurate and computationally efficient recognition compared to conventional graph convolutional methods.
4) **Extensive Evaluation:** We evaluate the proposed framework on four egocentric datasets. The results show that it outperforms state-of-the-art action recognition methods, demonstrating strong effectiveness, robustness, and scalability across modalities. Furthermore, our DEFT–DPT framework achieves 46.25% less computation time compared to existing methods, highlighting its efficient and lightweight design for fine-grained egocentric action recognition.

The rest of the paper is organized as follows. Section II reviews related work on action recognition. Section III introduces our proposed framework and discusses its computational complexity. Section IV presents experimental results along with a detailed analysis. Finally, Section V concludes the paper and outlines directions for future research.

II. Related Works

Egocentric perception has become a key research focus in computer vision and multimedia due to its wide range of applications in human action, object, and event recognition. We next review prior work on action recognition, covering both single-modality and multimodal approaches.

A. Single-Modal Egocentric Action Recognition

Early egocentric action recognition methods relies on hand-crafted features such as Histogram of Oriented Gradients

(HOG), Dense Trajectories, and SIFT to describe motion and appearance cues from first-person videos [5]. While effective at capturing local spatiotemporal information, these approaches struggle with strong ego-motion, frequent occlusions, and complex background variations. With the rise of deep neural learning, convolutional architectures such as C3D and I3D learn hierarchical spatiotemporal features directly from RGB sequences, yielding effective data-driven representations. However, these networks were originally designed for third-person videos, they often fail to capture egocentric-specific cues such as hand–object interactions, camera motion, and task-oriented attention patterns essential for wearable-view understanding.

To address these limitations, recent research has focused on explicitly modeling the egocentric characteristics. Sahu and Chowdhury [14] propose a weakly supervised superpixel-level framework for recognition, localization, and summarization using a Center-Surround Model and a weighted video graph with random-walk-based labeling, but overlook multimodal cues such as motion, audio, and object interactions. Attention-based models highlight key interaction regions and prioritize frames with meaningful manipulation cues [15], while other work models hand–object contact and object-state variations to improve temporal understanding [16]. Skeleton-based methods such as 2D^3-SkelAct [17] decouple and distill 3D latent features from 2D skeletons for robust recognition under viewpoint and depth ambiguity, but discard RGB and object-context cues vital for egocentric hand–object interactions. Zone-aware representations further improve generalization across varying environments, and large-scale datasets such as GTEA [18] and ADL [19]. This has strengthened the single-modality research through more diverse video collections. Lightweight pipelines such as Dark-DSAR [20] address adverse imaging conditions via compact one-step illumination adaptation, yet remain confined to third-person settings and overlook egocentric hand–object dynamics. Despite this progress, occlusion, lighting variation, and lack of depth information continue to limit RGB-only models, motivating a shift toward multimodal egocentric action recognition that integrates complementary cues such as depth, optical flow, gaze, or audio with RGB for richer, context-aware representations.

B. Multimodal Egocentric Action Recognition

To address the limitations of RGB-only input, many studies have explored multimodal approaches for egocentric action recognition. Optical flow is commonly used to capture short-term motion and subtle hand–object interactions that are difficult to infer from appearance alone. The two-stream architecture [9], which processes RGB and optical flow separately, has been extended with fusion strategies such as feature concatenation, attention weighting, and bilinear pooling to better combine spatial and temporal cues in dynamic egocentric scenarios.

In complex indoor environments, depth information provides useful 3D spatial context between the wearer and surrounding objects. Datasets such as MECCANO [21] have encouraged RGB-D models using 3D CNNs and transformer variants. Other cues, including gaze direction, audio, and hand segmentation masks, have also been explored for intent prediction and early action anticipation [22], [23]. Multimodal few-shot frameworks such as MM-CDFSL [24] combine RGB, optical flow, and hand-pose cues through distillation, but depend on static teacher–student learning and fixed detectors, limiting adaptive spatial reasoning. However, most multimodal frameworks [10], [25], [26] still use static or early-fusion schemes that fail to capture the changing importance of frames and modalities during egocentric interactions.

Recent works have advanced multimodal understanding through transformer and vision–language paradigms. Methods such as TransCLIP [27] and language-assisted representation learning [28] demonstrate the benefits of cross-modal alignment, while modular neural motion retargeting [29] and latent 3D feature decoupling [17] enhance motion and skeleton-based reasoning. Adaptive attention-based domain adaptation methods such as UDA-EAR [30] employ dual-branch attention with adversarial alignment for cross-domain egocentric recognition, yet incur extra computational overhead and lack explicit hand–object spatial transformation and scalable graph-based reasoning. Despite this progress, no framework yet captures the full spatial, temporal, and multimodal complexity of egocentric actions. To address this gap, our model integrates Graph Attention Networks (GATs) with a multimodal fusion strategy combining RGB and optical flow features. The Dynamic Egocentric Feature Transformation (DEFT) module learns adaptive spatial transformations that focus on salient hand–object regions while preserving viewpoint-specific cues essential for robust recognition. Unlike existing models that struggle with scalability on long or high-resolution sequences, our sparse representation ensures improved computational efficiency, and the resulting graph is processed by a GAT for context-aware feature aggregation, yielding compact yet discriminative representations for egocentric action recognition.

III. Proposed Method

The proposed framework, illustrated in the block diagram of Figure 2, comprises four main components: (a) Dynamic Egocentric Feature Transformation (DEFT), (b) feature extraction, (c) Sparse Video Similarity Graph (SVSG), and (d) Graph Attention Network (GAT). We next describe each component of the solution in detail.

Initially, the framework selects the available modalities RGB, motion (optical flow), and depth according to each dataset, ensuring that complementary information is captured for subsequent processing. Let a video X consist of a set of N frames, $X = \{x_1, x_2, \cdots, x_N\}$, where each frame x_i contains features from the available modalities. In the first component, these selected modalities are fed into the Dynamic Egocentric Feature Transformation (DEFT) module, which adaptively encodes the frame-level features while preserving the unique motion and viewpoint characteristics of egocentric videos.

A. Dynamic Egocentric Feature Transformation (DEFT)

Egocentric videos offer distinct advantages [6]: (a) important activities, actions, events, and objects are often located near the center of the frame, and (b) they are rarely occluded.

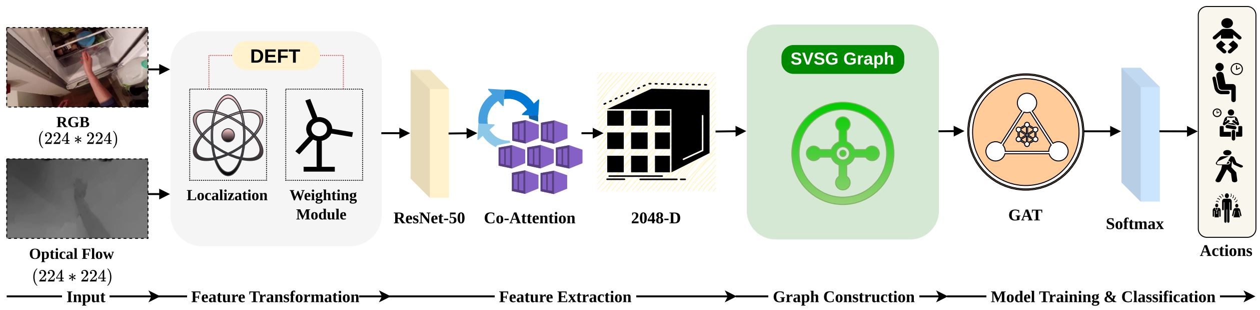


Fig. 2. A flowchart of the proposed framework.

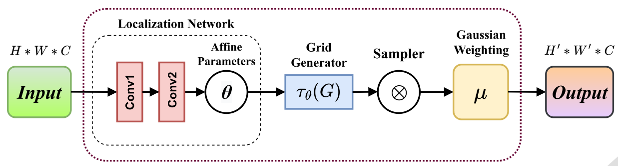


Fig. 3. Block diagram of dynamic egocentric feature transformation (DEFT).

Effectively capturing these spatial and motion cues is essential for accurate recognition of actions and objects. To address this, we introduce the Dynamic Egocentric Feature Transformation (DEFT) module, which adaptively encodes features while preserving the distinctive motion and viewpoint characteristics of egocentric videos. DEFT enhances spatial representations by focusing on informative regions within each frame and consists of two main components: a Localization Network (STN) [4] and a Gaussian Weighting Module. The STN learns affine transformation parameters to align feature maps toward action-relevant regions, such as hands or manipulated objects, reducing variations caused by ego-motion. The Gaussian Weighting Module applies an egocentric prior through a Gaussian-based spatial weighting function. Importantly, this weighting is applied in the affine-transformed space produced by the STN, implies that the spatial focus adapts per-frame: when the STN shifts hand-centric content to a non-central image location, the Gaussian peak follows accordingly. Together, these components enable viewpoint-invariant, dynamically adaptive feature extraction, improving the recognition of actions and objects in egocentric videos. Building on the detailed block diagram of the DEFT module in Figure 3, we next explain how DEFT adaptively encodes features from input video frames, preserving egocentric motion and viewpoint characteristics for accurate action recognition.

The DEFT module begins with a lightweight Spatial Transformer Network (STN) [4] that predicts affine transformation parameters to localize action-relevant regions, such as hands or interacted objects. The STN estimates the affine transformation matrix θ that aligns the input feature map into a canonical reference space. The network comprises two convolutional layers followed by max pooling and fully connected layers. The first convolutional layer applies 8 filters of size 7×7, followed by 2×2 max pooling; the second layer uses 10 filters of size 5×5, again followed by 2×2 pooling. The flattened output is passed through a fully connected layer with 32 hidden units and a final fully connected layer with 6 units to generate the affine transformation parameters:

$$\theta = \begin{pmatrix} \theta_1 & \theta_2 & \theta_3 \\ \theta_4 & \theta_5 & \theta_6 \end{pmatrix} \tag{1}$$

The transformation maps the original coordinates (u, v) to new coordinates (u', v'):

$$T(u, v) = \begin{pmatrix} u' \\ v' \end{pmatrix} = \begin{pmatrix} \theta_1 u + \theta_2 v + \theta_3 \\ \theta_4 u + \theta_5 v + \theta_6 \end{pmatrix} \tag{2}$$

where, θ_1, θ_2, θ_4, and θ_5 control scaling, rotation, and shearing, while θ_3 and θ_6 represent translations along the x and y axes. After transformation, bilinear interpolation ensures smooth resampling:

$$\mathcal{V}(u', v') = \sum_{i,j} \mathcal{V}(i, j) \cdot (1 - |u' - i|) \cdot (1 - |v' - j|) \tag{3}$$

where, $\mathcal{V}(i, j)$ denotes the feature value at integer pixel (i, j), and the interpolation weights enable sub-pixel precision.

To learn spatial alignment without explicit ground-truth transformation labels, DEFT uses a self-supervised alignment strategy. For each input frame $\mathbf{x}$, the module learns an affine transformation to produce a spatially aligned output $\mathbf{x}_{\text{transformed}}$ that emphasizes hand-centric content. The alignment target $\mathbf{x}_{\text{target}}$ is generated automatically under the assumption that hand–object interactions are usually centered in egocentric frames. Instead of fixed cropping, $\mathbf{x}_{\text{target}}$ corresponds to the affine-transformed frame without weighting, serving as a weak reference for alignment. Both share the same affine parameters, differing only on whether weighting is applied. The alignment loss is defined as:

$$\mathcal{L}_{\text{align}} = \frac{1}{N} \sum_{i=1}^{N} \left\| x_{\text{transformed}}^{(i)} - x_{\text{target}}^{(i)} \right\|_2^2 \tag{4}$$

where, N is the number of training samples. This loss enforces consistency between the transformed and target frames, stabilizing training and preserving egocentric alignment.

Examples of DEFT's transformations are shown in Figure 4, illustrating (a) the original egocentric frame, (b) the alignment target, (c) the affine-transformed output, and (d) the final Gaussian-weighted output $\mathbf{x}_{\text{DEFT}}$. These visualizations demonstrate how DEFT progressively concentrates

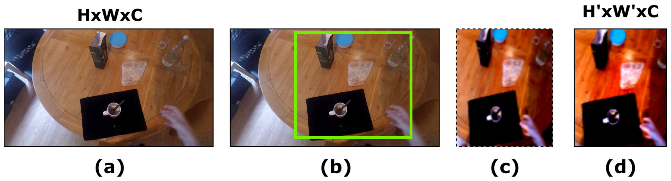


Fig. 4. Visualization of DEFT processing on an egocentric frame: (a) original frame, (b) reference crop for alignment, (c) spatially transformed output, (d) final weighted feature map $\mathbf{x}_{\text{DEFT}}$ after Gaussian weighting.

on hand–object interaction regions. To further refine spatial focus, DEFT applies a dynamic Gaussian weighting function that enhances attention toward the most relevant regions in each frame. Unlike fixed-center cropping, which assumes static hand–object positions, this adaptive approach learns frame-wise spatial attention guided by egocentric priors. The Gaussian weighting function is defined as:

$$w(u',v') = 1 + \lambda \exp\left(-\frac{\|T(u,v)-\mathbf{c}\|_2^2}{2\sigma^2}\right) \tag{5}$$

where, (u',v') represents the pixel location, λ controls attention intensity, σ defines spatial spread, and $\mathbf{c} = (u_c, v_c)$ indicates the attention center. The center $\mathbf{c}$ is defined in the canonical reference space, while the Gaussian weighting is evaluated on the transformed coordinates $(u',v') = T(u,v)$, where the affine transformation T is predicted per-frame by the Spatial Transformer Network (Eqs. (1)–(2)). Through this formulation, the spatial peak of $w(u',v')$ shifts across frames according to the predicted affine parameters, allowing DEFT to attend to off-center hand–object interactions by mapping the same canonical center $\mathbf{c}$ to different image-space locations from frame to frame. To prevent over-concentration on small areas, a regularization term maintains weight uniformity:

$$\mathcal{L}_{\text{weighting}} = \frac{1}{N}\sum_{t=1}^{N}\left\|w(u',v')-1\right\|_2^2 \tag{6}$$

This regularization stabilizes attention learning and ensures spatial generalization. To effectively capture both geometric and contextual variations, the final DEFT representation combines spatial transformation with adaptive weighting before being optimized using a multi-task loss:

$$\mathcal{F}_{\text{DEFT}}(u',v') = w(u',v')\cdot\mathcal{F}(\mathcal{X})(T(u,v)) \tag{7}$$

where, $\mathcal{F}(\mathcal{X})$ is the feature map extracted from input $\mathcal{X}$, transformed by $T(u,v)$ and scaled by learned weight $w(u',v')$. DEFT is optimized using a multi-task loss:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{align}} + \eta\mathcal{L}_{\text{reg}} + \kappa\mathcal{L}_{\text{weighting}} \tag{8}$$

Here, the regularization term is defined as:

$$\mathcal{L}_{\text{reg}} = \psi\|\theta - I\|_2^2 \tag{9}$$

where, I is the identity matrix and θ the predicted affine matrix. This constraint the transformations close to identity unless major deformations are needed. Scalars η, ψ, and κ control each loss term's contribution, ensuring stable training and convergence.

Note that without DEFT, using fixed cropping reduces the F1-Score by 3.7%, highlighting the advantage of adaptive feature weighting through the DEFT module.

B. Feature Extraction

After processing through DEFT, we employ deep learning in an unsupervised manner to extract visual descriptors from the central region of each video frame. Specifically, a pre-trained ResNet-50 [31] is used owing to its strong generalization capability and proven effectiveness in visual representation learning. The network is composed of convolutional layers and residual connections that enable efficient feature propagation and gradient flow. For each frame, the deep feature descriptor is obtained from the global average pooling layer, yielding a 2048-dimensional vector representation of the central region:

$$f_i = \phi(x_{DEFT\,i}) \in \mathbb{R}^{2048}, \tag{10}$$

where, $x_{DEFT\,i}$ denotes the i^{th} input frame and $\phi(\cdot)$ represents the feature extraction function of ResNet-50. This single-stream formulation extends naturally to multiple modalities. When multiple modalities are available, $\phi(\cdot)$ is applied independently to each stream, producing per-frame descriptors $f_i^{(r)}$ and $f_i^{(o)} \in \mathbb{R}^{2048}$ for RGB and optical flow, respectively. To integrate these complementary cues, a lightweight Co-Attention adaptively reweights each stream based on its agreement with the other. Given the ℓ_2-normalized descriptors $\hat{f}_i^{(r)}$ and $\hat{f}_i^{(o)}$, a scalar agreement coefficient is defined as

$$\alpha_i = \sigma\left(\frac{\left(\hat{f}_i^{(r)}\right)^{\top}\hat{f}_i^{(o)}}{\nu}\right), \tag{11}$$

where, $\sigma(\cdot)$ is the sigmoid function and ν is a temperature controlling the gating sharpness. Each modality is then enhanced by a residual contribution from the other, modulated by α_i:

$$\tilde{f}_i^{(r)} = f_i^{(r)} + \alpha_i f_i^{(o)}, \quad \tilde{f}_i^{(o)} = f_i^{(o)} + \alpha_i f_i^{(r)}. \tag{12}$$

The two attended representations are concatenated and projected back to the original 2048-dimensional space via a learned linear layer $W_p \in \mathbb{R}^{2048\times4096}$, yielding the fused feature descriptor used in subsequent stages:

$$f_i = W_p\left[\tilde{f}_i^{(r)};\, \tilde{f}_i^{(o)}\right] \in \mathbb{R}^{2048} \tag{13}$$

C. Sparse Video Similarity Graph

We construct a weighted Sparse Video Similarity Graph (SVSG), represented as $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, in the deep feature space. The sparsity of $\mathcal{G}$ offers distinct advantages. First, operating the Graph Attention Network (GAT) on a sparse graph considerably lowers computational complexity compared to dense or fully connected graphs. Moreover, the sparse structure preserves semantic relevance, as edges are established only between spatially and temporally adjacent frames, ensuring efficient and meaningful feature propagation across the video sequence.

Initially, we construct a weighted complete graph, referred to as the *Video Similarity Graph (VSG)*, in a 2048-dimensional feature space to represent all frames of a video. Each frame serves as a vertex, and every pair of vertices is fully connected.

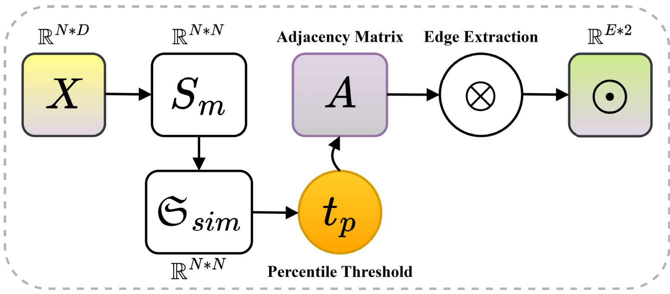


Fig. 5. Block diagram of SVSG graph construction using DPT.

The edge weight S_{ij} between vertices i and j, associated with feature vectors $f_i, f_j \in \mathbb{R}^{2048}$, is computed using cosine similarity as follows:

$$S_{ij} = \frac{f_i f_j}{\|f_i\|_2 \, \|f_j\|_2}, \quad \forall\, i, j \in \mathcal{V}. \tag{14}$$

This similarity defines the connection strength between frames, capturing their semantic and visual correspondence in the deep feature space. To construct the sparse graph, we employ a Dynamic Percentile Threshold (DPT) strategy that adaptively prunes edges based on similarity values derived from the feature distribution. Unlike conventional fixed k-nearest neighbor approaches, DPT retains only those node pairs whose similarity scores exceed a dynamically computed percentile threshold. This adaptive mechanism ensures that connections are established only between frames exhibiting strong contextual similarity, thereby producing a sparse yet semantically coherent graph that better reflects temporal and spatial relationships within the video. The detailed block diagram of the SVSG graph construction module is illustrated in Figure 5. As shown, the input feature matrix $\mathcal{X} \in \mathbb{R}^{N\times D}$, where N denotes the number of nodes and D represents the feature dimension, is first processed to compute a similarity matrix $\mathcal{S}_m \in \mathbb{R}^{N\times N}$, capturing pairwise cosine similarities among all nodes. Based on these similarity values, an adjacency matrix $A \in \{0,1\}^{N\times N}$ is constructed by retaining only those node pairs whose similarity exceeds a dynamically computed percentile threshold, thereby forming an adaptively sparse and semantically meaningful graph structure. We next describe the construction of the adaptively sparse video similarity graph (SVSG) using Dynamic Percentile Thresholding (DPT). Further, to retain only the most informative edges and reduce graph density, we apply a *Dynamic Percentile Threshold (DPT)* strategy. The percentile threshold value t_p is computed as the p^{th} percentile of all similarity values in $\mathcal{S}_m$:

$$t_p = \Xi(\mathcal{S}_m, p) \tag{15}$$

In practice, this threshold is computed in three steps:

$$\begin{aligned}
&\text{Flatten similarity matrix:} \quad \mathfrak{F}_{\text{sim}} = \boxminus(\mathcal{S}_m)\\
&\text{Sort similarity values:} \quad \mathfrak{S}_{\text{sim}} = \circledS(\mathfrak{F}_{\text{sim}})\\
&\text{Compute threshold:} \quad t_p = \mathfrak{S}_{\text{sim}}\left[\left\lceil \frac{p\cdot\mathcal{L}(\mathfrak{S}_{\text{sim}})}{100}\right\rceil\right]
\end{aligned} \tag{16}$$

where, $\boxminus$ denotes the flattening operation, $\circledS$ represents sorting in ascending order, and $\mathcal{L}(\cdot)$ indicates the vector length. The resulting threshold t_p ensures that only the top $(100-p)\%$ of similarity values are retained, effectively filtering out weak contextual connections.

Once the threshold is determined, edges corresponding to node pairs that satisfy $\mathcal{S}(i,j) \geq t_p$ are retained. The resulting set of edges is defined as:

$$\odot = \{(i,j) \mid \mathcal{S}(i,j) \geq t_p,\ i \neq j\} \tag{17}$$

where, $\odot \in \mathbb{R}^{\mathcal{E}\times 2}$ and $\mathcal{E}$ denotes the total number of retained edges. Each row in $\odot$ corresponds to a connected node pair (i,j), forming the final sparse graph structure that serves as the input to the downstream GNN.

This dynamic sparsification preserves the most meaningful feature relationships in egocentric videos. By using percentile-based thresholds instead of fixed neighbors, DPT creates a context-sensitive, computationally efficient graph that enhances temporal and semantic reasoning. The resulting adaptive graph is then passed to a Graph Attention Network (GAT), which models temporal dependencies and frame-wise contextual importance through attention-based message propagation.

D. Graph Attention Network (GAT)

A Graph Attention Network (GAT) is employed to capture temporal dependencies and contextual relationships among frames, predicting action labels for each node through attention-weighted message passing. The resulting sparse video similarity graph is represented as $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where V denotes the set of nodes, and $\mathcal{E} \in \mathbb{R}^{2\times \text{num\_edges}}$ represents the dynamically connected edges, defining relationships between nodes. The GAT model processes the input through multiple layers to refine the embeddings. The first GAT layer applies a graph convolution with multiple attention heads, followed by an ELU activation:

$$\mathbf{H}^{(1)} = \Vert_{k=1}^{K}\, \sigma\left(\sum_{j\in\mathcal{N}_i} \alpha_{ij}^{k} \mathbf{W}^{k} \mathbf{h}_j\right) \tag{18}$$

where, K is the number of attention heads, and σ is the non-linear activation function, typically ELU. The output from this layer has the shape$[N \times (H \times \text{heads})]$

The second GAT layer further refines the node embeddings and is given by:

$$\mathbf{H}^{(2)} = \sigma\left(\sum_{j\in\mathcal{N}_i} \alpha_{ij} \mathbf{W} \mathbf{h}_j^{(1)}\right) \tag{19}$$

This layer outputs a matrix of shape $[N \times C]$, where, C is the number of output classes. To classify the nodes, a softmax function is applied to the output of the second GAT layer:

$$\hat{Y}_{i,c} = \frac{\exp\left(H_{i,c}^{(2)}\right)}{\sum_{c'=1}^{C} \exp\left(H_{i,c'}^{(2)}\right)} \tag{20}$$

This step maintains the same shape as the output of the second GAT layer i.e. $[N \times C]$, The final output layer assigns a

Algorithm 1 DEFT–DPT Framework for Egocentric Action Recognition

Input: An egocentric video sequence (from user)
Output: Action localization within each frame and overall action recognition for the video

Step 1: Input and Modality Selection:
Initially, an input video $X = \{x_1, x_2, \ldots, x_N\}$ is considered. For each dataset, the available modalitiesâ"such as RGB, optical flow, or depthâ"are selected to serve as the input streams.

Step 2: Localization (Spatial Alignment):
Use an STN to learn affine transformations that align action-relevant regions, reducing ego-motion effects.

Step 3: Adaptive Spatial Weighting (DEFT Weighting):
Apply Gaussian-based spatial weighting to emphasize informative central areas and suppress background noise.

Step 4: Deep Feature Extraction:
Extract high-level feature embeddings from the weighted frames of each modality using a CNN backbone.

Step 5: Co-Attention Fusion:
Compute a per-frame agreement coefficient between modality descriptors and apply cross-modal residual updates to generate fused frame representations.

Step 6: Sparse Video Similarity Graph (SVSG) Construction:
Compute pairwise cosine similarities between frame features to build a dense similarity graph. Apply Dynamic Percentile Thresholding to retain significant connections and obtain a sparse graph.

Step 7: Graph Learning with Graph Attention Network (GAT):
Process the sparse graph using a GAT to model temporal dependencies and contextual relationships through attention-weighted message passing.

Step 8: Classification and Output:
Predict frame-level action labels based on the learned graph representations.

predicted class to each node by selecting the class with the highest probability:

$$\text{Class}(i) = \arg\max_{c \in C} \hat{Y}_{i,c} \tag{21}$$

We conclude this section with Algorithm 1, which summarizes the complete sequence of steps in the proposed method described below.

E. Time-Complexity Analysis

We now analyze the computational complexity of the proposed *DEFT–DPT framework*. Feature extraction using a CNN backbone for N frames with a kernel size of $K \times K$ incurs a cost of $\mathcal{O}(N \cdot D \cdot K^2)$, where D is the feature dimension. The Localization Network (STN) predicts affine transformation parameters and performs coordinate transformation with a complexity of $\mathcal{O}(N \cdot D)$, while the Dynamic Weighting Module adds $\mathcal{O}(N)$. Thus, the overall complexity of the DEFT module is approximated as: $\mathcal{O}(N \cdot D \cdot K^2) + \mathcal{O}(N \cdot D) + \mathcal{O}(N) \approx \mathcal{O}(N \cdot D \cdot K^2)$. The Co-Attention adds a per-frame cost of $\mathcal{O}(D \cdot D')$, dominated by the linear projection, yielding a total of $\mathcal{O}(N \cdot D \cdot D')$, which is linear in N. For sparse graph construction using DPT, pairwise cosine similarity computation contributes $\mathcal{O}(N^2 \cdot D)$, followed by sorting for percentile thresholding with $\mathcal{O}(N^2 \log N)$, and edge selection with $\mathcal{O}(|E|)$. Hence, the total complexity of the SVSG process is $\mathcal{O}(N^2 \cdot D + N^2 \log N + |E|)$. Combining both components, the overall computational complexity of the proposed DEFT–DPT pipeline can be approximated as: $\mathcal{O}(N \cdot D \cdot K^2 + N \cdot D \cdot D' + N^2 \cdot D + N^2 \log N + |E|) \approx \mathcal{O}(N^2 \cdot D) \approx \mathcal{O}(N^2)$. For very long video sequences ($N > 500$), the framework can be deployed via a sliding-window strategy with fixed window size $W \ll N$, reducing the SVSG complexity from $\mathcal{O}(N^2)$ to $\mathcal{O}(N \cdot W)$–linear in N.

IV. Experimental Results

All experiments were conducted on a workstation with an Intel Xeon W-2295 CPU (18 cores, 3.00 GHz), 128 GB DDR4 RAM, and an NVIDIA RTX A4000 GPU. We first describe the datasets, performance measures, and implementation details, followed by hyperparameter analysis and parameter settings. Next, ablation studies are presented to validate the contribution of each framework component. We then compare the proposed method with state-of-the-art egocentric action recognition approaches, including action localization analysis, foundation model comparisons, and computational efficiency evaluation. Finally, we provide a Graph Connectivity and Sparsity Analysis.

A. Datasets and Performance Measures

We use four egocentric video datasets for our experiments: EPIC-Kitchens [9], Meccano [21], GTEA [18], and ADL [19]. For performance measures, For performance measures, we employ Top-1 and Top-5 Accuracy, and F-score as evaluation metrics. Detailed information about these datasets and performance evaluation is provided in the supplementary material. AQ:3

B. Implementation Details

The framework was implemented using PyTorch and PyTorch Geometric with GPU acceleration. For data preprocessing, video frames from available datasets were uniformly sampled with a stride of $S = 10$, resized to 224×224, converted to tensors, and normalized using ImageNet statistics (mean $= [0.485, 0.456, 0.406]$, std $= [0.229, 0.224, 0.225]$). In the EPIC-Kitchens dataset, optical flow frames computed using the TV-L1 algorithm were used directly and aligned with RGB frames for ResNet-50 feature extraction. Finally, the GAT model was trained using the Adam optimizer (θ) with a learning rate and weight decay of 1×10^{-4} for 100 epochs, while the DEFT module was optimized using a multi-task loss ($\alpha = 0.8$) and Gaussian regularization ($\sigma = 0.5$).

C. Hyperparameter Sensitivity Analysis

To identify the optimal training configuration, we conducted a hyperparameter sensitivity analysis on the EPIC-Kitchens

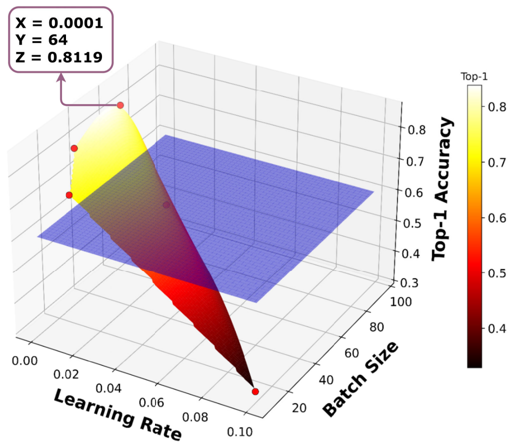


Fig. 6. Hyperparameter Sensitivity: Top-1 accuracy on EPIC-Kitchens P01_04 video, peaks at a learning rate lr = 0.0001 and batch size B = 64, balancing convergence and stability.

TABLE I

HYPERPARAMETER DETAILS OF THE PROPOSED FRAMEWORK

Parameter	Value
Optimizer (θ)	Adam
Batch Size (B)	64
Weight Decay (β)	1×10^{-4}
Learning Rate (lr)	1×10^{-4}
Number of Epochs (E)	100
Co-Attention Temperature (ν)	0.1

dataset, examining the impact of different learning rates ($1 \times 10^{-5} - 1 \times 10^{-1}$) and batch sizes ($16 - 64$). As shown in Figure 6, variations in these two hyperparameters significantly influence performance. We observe notable differences in Top-1 accuracy across different configurations, with a learning rate of 1×10^{-4} (lr) and a batch size of 64 (B) achieves the highest Top-1 accuracy of 81.19%. This configuration outperforms both higher learning rates (e.g., 0.01 and 0.1), which cause unstable training. The smaller learning rates like 0.00001, results in slower convergence. Finally, the hyperparameters were tuned during training, and the final values are summarized in Table I.

D. Parameter Setting

We evaluated the dynamic graph construction parameter (τ) through a sensitivity analysis across percentile values ranging from 90 to 99, as shown in Figure 7. Experiments on representative videos from four benchmark egocentric datasets reveal that the thresholds between 94 and 96 percentiles achieve the best F-score. The performance saturates beyond threshold of 97 and declines below 94, indicating that moderately high thresholds effectively capture salient temporal interactions while suppressing redundant connections.

E. Ablation Studies

We analyze the contributions of each component in our framework to demonstrate their individual impact on

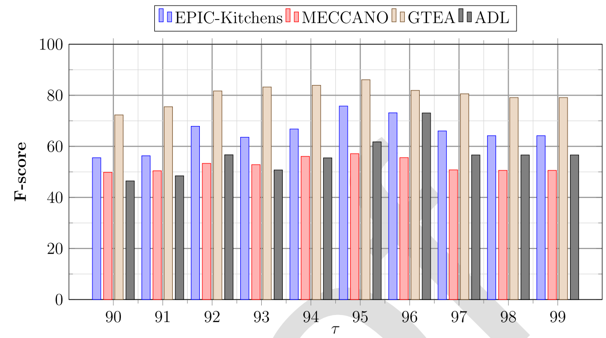


Fig. 7. F-Score for different values of τ on P01_04 video from Epic-Kitchens, 0005 video from Meccano, S2_Cheese_C1 video from GTEA and P_16 video from ADL datasets.

TABLE II

ABLATION STUDY 1: IMPACT OF DEFT VS. DIRECT FRAME & STN

Methods	**EPIC-Kitchens** [9]	**Meccano** [21]	**GTEA** [18]
$P + R + S + T$	0.6449	0.6157	0.9030
$P + Q_1 + R + S + T$	0.6828	0.6953	0.9299
$P + Q + R + S + T$	**0.7049**	**0.7093**	**0.9319**

*(P = Input Frames, R = ResNet-50, S = DPT, Q_1 = Localization (STN), Q = DEFT, T = GAT)

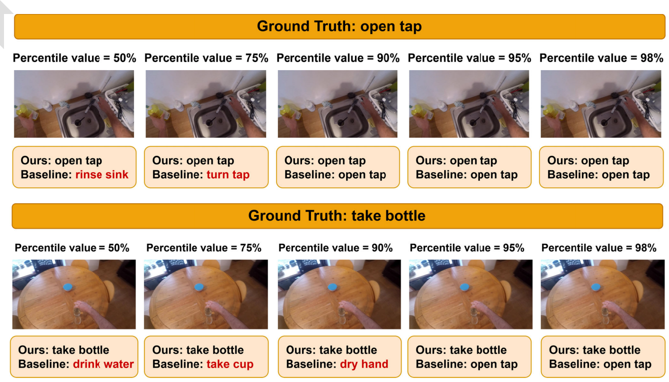


Fig. 8. Action predictions of our model with and without DEFT (denoted as the baseline) over percentile value with two testing examples.

overall performance, including: (i) Dynamic Egocentric Feature Transformation (DEFT), (ii) impact of Concatenation vs. Co-Attention Fusion, (iii) the impact of individual loss components in DEFT, (iv) the impact of different feature extraction models, (v) the influence of various sparse graph construction methods, and (vi) the role of the Graph Attention Network (GAT). In the first ablation study, we evaluate the improvements gained by using DEFT compared to a direct frame-level approach without DEFT. When a basic localization step using a Spatial Transformer Network (STN) is added, the results improve moderately, confirming that spatial alignment helps stabilize hand-centric representations. The mean accuracy values for action recognition in Table II clearly demonstrate the effectiveness of incorporating DEFT.

In addition, the qualitative results in Figure. 8., clearly shows that our model with DEFT correctly identifies actions such as *"open tap"* and *"take bottle"*, whereas the baseline

TABLE III

ABLATION STUDY 2: IMPACT OF CONCATENATION VS. CO-ATTENTION FUSION

Methods	EPIC-Kitchens [9]	Meccano [21]	GTEA [18]
$P + Q + R + F_{\{concat.\}} + S + T$	**0.7012**	**0.7037**	**0.9314**
$P + Q + R + F_{\{co-att.\}} + S + T$	**0.7049**	**0.7093**	**0.9319**

*(P = Input Frames, Q = DEFT, R = ResNet-50, $F_{\{concat.\}}$ = Concatenation, $F_{\{co-att.\}}$ = Co-Attention, S = DPT, T = GAT)

TABLE IV

ABLATION STUDY 3: IMPACT OF LOSS COMPONENTS AND THEIR COMBINATIONS ON DEFT PERFORMANCE

Methods	EPIC-Kitchens [9]	Meccano [21]	GTEA [18]
ζ_1	0.6720	0.6853	0.8827
ζ_2	0.6625	0.6777	0.8762
ζ_3	0.6592	0.6725	0.8714
$\zeta_1 + \zeta_2$	0.6848	0.6927	0.8889
$\zeta_1 + \zeta_3$	0.6877	0.6945	0.8897
$\zeta_2 + \zeta_3$	0.6816	0.6898	0.8865
$\zeta_4 = \zeta_1 + \zeta_2 + \zeta_3$	0.6924	0.7024	0.9020
$\zeta_5 = L_{\mathrm{CE}} + \zeta_4$	0.6877	0.6580	0.8930
$\zeta_6 = \zeta_5 + L_{\mathrm{temp}}$	**0.7049**	**0.7093**	**0.9319**

model without DEFT often confuses them with visually similar actions like *"turn tap"* or *"drink water"*.

In the second ablation study, we evaluated the impact of the cross-modal fusion strategy on action recognition performance. Specifically, we compared two configurations of the proposed framework: (i) simple feature concatenation ($F_{\{concat.\}}$), where the per-modality descriptors from RGB and optical flow are directly concatenated before being passed to subsequent stages, and (ii) the proposed Co-Attention fusion ($F_{\{co\text{-}att.\}}$), where each modality is adaptively reweighted based on its agreement with the other prior to aggregation. The results in Table III clearly show that replacing simple concatenation with the proposed Co-Attention consistently improves Top-1 accuracy across all three datasets, highlighting the effectiveness of agreement-based modality reweighting in capturing complementary spatial and motion cues for egocentric action recognition.

In the third ablation study, we evaluated the impact of each loss term on action recognition performance. Specifically, we considered the DEFT-related spatial losses $\zeta_1 = L_{\mathrm{align}}, \zeta_2 = L_{\mathrm{weighting}}, \zeta_3 = L_{\mathrm{reg}}$ as well as various combinations of these components (e.g., $\zeta_1 + \zeta_2$, $\zeta_1 + \zeta_3$, $\zeta_2 + \zeta_3$, $\zeta_4 = \zeta_1 + \zeta_2 + \zeta_3$) to assess their effect on local feature alignment and stability. Additionally, global objectives including Cross-Entropy (L_{CE}) and Temporal Consistency (L_{temp}) were used to guide discriminative and temporally coherent learning. The results in Table IV clearly shows that the proposed combined loss function outperforms individual loss terms or partial combinations, highlighting its effectiveness in improving DEFT-based action recognition accuracy.

In the fourth ablation study, we retained DEFT, DPT, and GAT while varying the CNN-based feature extractors. Specifically, we compared action recognition performance using different CNN-based feature extraction pre-trained networks. As shown in Table V, ResNet-50 consistently achieves the best overall performance, demonstrating its superior ability to extract discriminative spatio-temporal features crucial for the DEFT–DPT framework.

TABLE V

ABLATION STUDY 4: IMPACT OF DIFFERENT FEATURE EXTRACTION METHODS

Methods	EPIC-Kitchens [9]	Meccano [21]	GTEA [18]
$P + Q + R_1 + S + T$	0.6931	0.6823	0.8936
$P + Q + R_2 + S + T$	0.6838	0.6625	0.8732
$P + Q + R_3 + S + T$	0.6939	0.6958	0.8925
$P + Q + R_4 + S + T$	0.7010	0.7004	0.8801
$P + Q + R_5 + S + T$	0.7037	0.7014	0.9317
$P + Q + R + S + T$	**0.7049**	**0.7093**	**0.9319**

*(P = Input Frames, Q = DEFT, R_1 = VGG16, R_2 = AlexNet, R_3 = EfficientNet-B0, R_4 = MobileNetV3, R_5 = ConvNeXt, R = ResNet-50, S = DPT, T = GAT)

TABLE VI

ABLATION STUDY 5: IMPACT OF DIFFERENT SPARSE GRAPH CONSTRUCTION METHODS

Methods	EPIC-Kitchens [9]	Meccano [21]	GTEA [18]
$P + Q + R + S_1 + T$	0.6440	0.6675	0.7758
$P + Q + R + S_2 + T$	0.6780	0.6730	0.8026
$P + Q + R + S_3 + T$	0.6832	0.6798	0.8129
$P + Q + R + S + T$	**0.7049**	**0.7093**	**0.9319**

*(P = Input Frames, Q = DEFT, R = ResNet-50, S_1 = K-NN, S_2 = GraphSAGE[32], S_3 = GIN[33], S = DPT, T = GAT)

TABLE VII

ABLATION STUDY 6: IMPACT OF GAT VS. GCN

Methods	EPIC-Kitchens [9]	Meccano [21]	GTEA [18]
$P + Q + R + S + T_1$	0.6142	0.6394	0.8540
$P + Q + R + S + T$	**0.7049**	**0.7093**	**0.9319**

*(P = Input Frame, Q = DEFT, R = ResNet-50, S = DPT, T_1 = GCN, T = GAT)

In the fifth ablation study, we kept the DEFT and ResNet-50-based feature extraction unchanged but varied the sparse graph construction techniques, including traditional fixed k-NN, GraphSAGE-based [32], and GIN-based [33] approaches. While the GIN-based graph slightly improves performance, it still fails to effectively model dynamic inter-frame correlations. In contrast, the proposed Dynamic Percentile Thresholding (DPT) method achieves a notable improvement. The results in Table VI clearly indicate that DPT yields the most superior performance.

In the sixth ablation study, we evaluated the impact of the Graph Attention Network (GAT) by keeping the DEFT-based solution, ResNet-50 feature extraction, and DPT-based sparse graph construction fixed, while varying the graph-based classification model. Specifically, we compared action recognition performance using two different graph classification strategies—Graph Convolutional Network (GCN) and

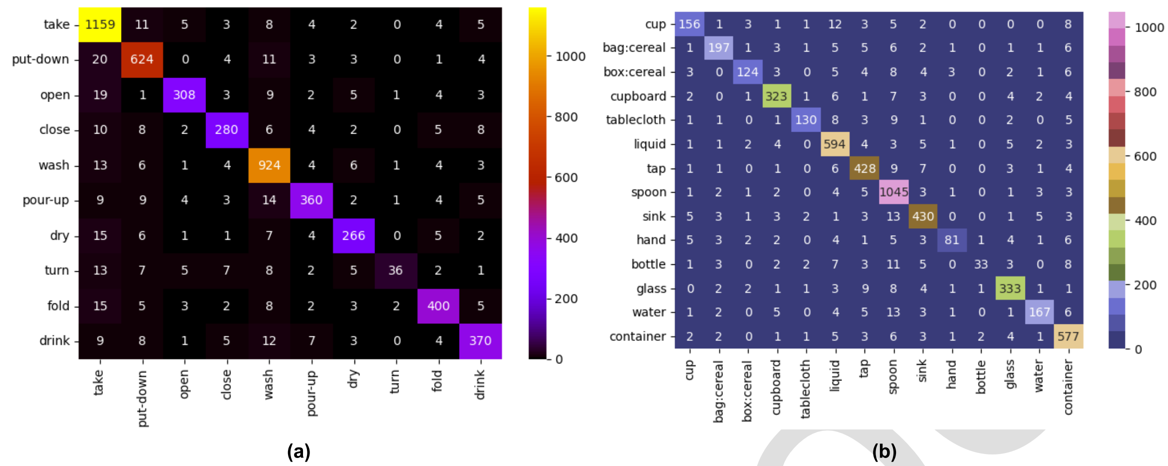


Fig. 9. Confusion matrix of our model for (a) 10 verb classes and (b) 14 noun classes on P01_04 video from the EPIC-Kitchens dataset.

TABLE VIII
COMPARISONS WITH STATE-OF-THE-ART METHODS ON THE EPIC-KITCHENS DATASET

Methods	Verb	Noun	Action
EAT-MBNet [31] (TCSVT' 2021)	0.6120	0.4340	0.3220
BB + KP + Ego-LSTM [3] (TPAMI' 2022)	0.6453	0.5576	0.4141
DeFormer [34] (TIP' 2023)	0.6710	0.5150	0.3970
SVFormer [2] (CVPR' 2023)	0.5960	–	0.4120
HOCL-OSL [16] (WACV' 2024)	0.5970	0.5263	0.3439
SMAST [7] (TPAMI' 2024)	0.7010	0.6480	0.5090
FFCN-OMNIVORE [8] (TCSVT' 2024)	0.7360	0.6170	0.6910
GPT4Ego [35] (TMM' 2024)	–	–	0.4260
TIM [36] (CVPR' 2024)	0.7620	0.6640	0.5640
IKD + SVFormer [1] (PR' 2025)	0.5220	–	0.4330
OURS	**0.7687**	**0.7194**	**0.7049**

Graph Attention Network (GAT). The results shown in Table VII. clearly demonstrate that GAT-based action recognition achieves superior performance.

F. Comparison With State-of-the-Art Methods

1) Results on the EPIC-Kitchens Dataset: We compare our proposed DEFT-DPT model with ten state-of-the-art methods, namely, IKD + SVFormer [1], SVFormer [2], EAT-MBNet [31], BB + KP + Ego-LSTM [3], HOCL-OSL [16], SMAST [7], DeFormer [34], FFCN-OMNIVORE [8], GPT4Ego [35], and TIM [36]. Table. VIII demonstrates that the proposed method consistently outperforms all these competing approaches across the evaluated benchmarks. Specifically, for Top-1 accuracy, our model attains 76.87%, 71.94%, and 70.49% for verb, noun, and action recognition, respectively, clearly demonstrating its effectiveness and robustness compared to existing approaches.

In addition, we present confusion matrices for verb-noun, and action classes in Figures 9, and 10, respectively. Comparing the ground-truth labels with the predicted labels, it is evident that our approach reduces inter-class confusion and produces more diagonalized confusion matrices across all categories.

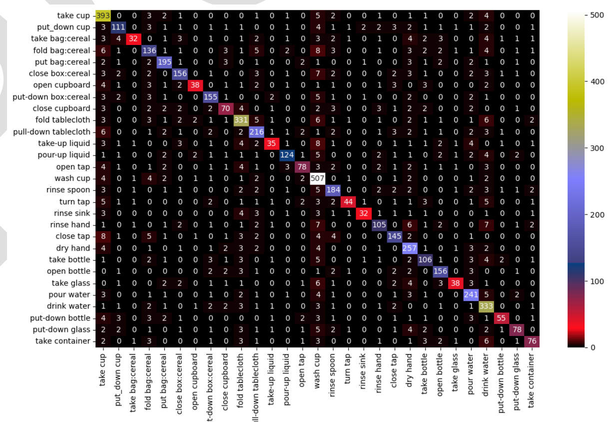


Fig. 10. Confusion matrix of our model for all 29 classes on P01_04 video from EPIC-Kitchens dataset.

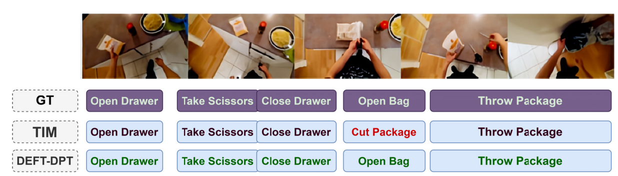


Fig. 11. Qualitative comparison between TIM [36] and our DEFT-DPT model on an egocentric sequence from the EPIC-Kitchens dataset.

Qualitative results: In Figure 11, five key actions names are *"Open Drawer"*, *"Take Scissors"*, *"Close Drawer"*, *"Open Bag"*, and *"Throw Package"* are highlighted. Our method correctly predicts all five actions, whereas the recent TIM [36] method misses one, misclassifying *"Open Bag"* due to limited spatial contextualization.

2) Results on the Meccano Dataset: We evaluate our method on the Meccano dataset against four state-of-the-art approaches: HOCL-OSL [16], UNICT [2], MACAU [26], and UCF [37]. As shown in Table IX, our approach outperforms all compared methods across Top-1 and Top-5 accuracy and

TABLE IX
COMPARISONS WITH STATE-OF-THE-ART METHODS ON THE MECCANO DATASET

Methods	Top-1	Top-5	F1-Score
UNICT [21] (WACV' 2021)	0.4949	0.7761	0.5190
MACAU [26] (PAMI' 2023)	0.5030	0.7846	–
UCF [37] (arXiv' 2023)	0.5537	0.8558	–
HOCL-OSL [16] (WACV' 2024)	0.4481	0.7701	0.1850
OURS	**0.7093**	**0.9536**	**0.5723**

TABLE X
COMPARISONS WITH STATE-OF-THE-ART METHODS ON THE GTEA DATASET

Methods	Top-1	Top-5	F1-Score
LSTA [15] (CVPR' 2019)	0.7931	–	–
DAN-EAR [38] (TIP' 2019)	0.4835	–	–
EAT-MBNet [31] (TCSVT' 2021)	0.8196	–	–
OURS	**0.9319**	**0.9909**	**0.7454**

TABLE XI
COMPARISONS WITH STATE-OF-THE-ART METHODS ON THE ADL DATASET

Methods	Top-1	Top-5	F1-Score
FPAR–DLD [39] (CVPR' 2016)	0.3758	–	–
EAP–EMA [40] (ECCV' 2018)	0.6190	–	–
MD–MTL [41] (PAMI' 2021)	0.6465	0.8838	–
OURS	**0.7599**	**0.9823**	**0.7313**

F1-score. Specifically, it achieves a Top-1 accuracy of 70.93% and an F1-score of 57.23%, demonstrating strong recognition performance and robust generalization across diverse egocentric actions.

3) Results on the GTEA Dataset: We compare the proposed method with three state-of-the-art approaches—DAN-EAR [38], EAT-MBNet [31], and LSTA [15]—using Top-1 and Top-5 accuracy, and F1-Score as evaluation metrics. As shown in Table X, our method achieves the highest performance, with Top-1 and Top-5 accuracy of 93.19% and 99.09% respectively, surpassing existing approaches. Additionally, it demonstrates superior Precision (77.99%), Recall (79.49%), and F1-Score (74.54%), underscoring its robustness in recognizing fine-grained egocentric actions.

4) Results on the ADL Dataset: We compare our proposed method with three state-of-the-art approaches, namely, FPAR–DLD [39], EAP–EMA [40], and MD–MTL [41], using Top-1 and Top-5 accuracy, as well as F1-Score as evaluation metrics. As shown in Table XI, our method achieves the best performance, with a Top-1 accuracy of 75.99% and Top-5 accuracy of 98.23%, significantly outperforming all prior methods. Additionally, our approach yields higher values in Precision (76.79%), Recall (75.99%), and F1-Score (73.13%), demonstrating its ability to maintain a strong balance across recognition accuracy.

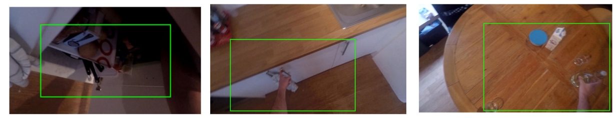


Fig. 12. Localization of simple actions in P01_03 video of the EPIC-Kitchens dataset: "take cereal" (label 25), "open cupboard" (label 0) and "move bottle" (label 11).

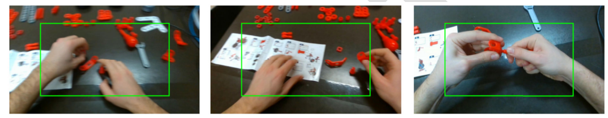


Fig. 13. Localization of simple actions in 0005 video of the Meccano dataset: "put bolt" (label 42), "take_red_perforated_junction_bar" (label 19) and "screw_screw_with_screwdriver" (label 26).

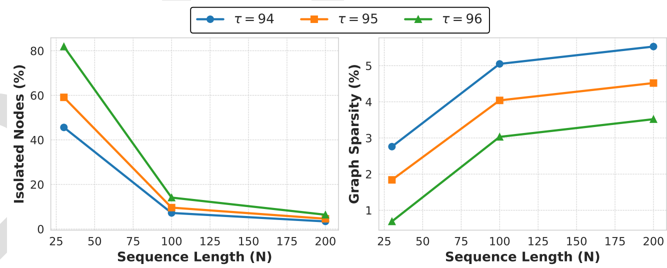


Fig. 14. Graph behavior of DPT at $\tau \in \{94, 95, 96\}$: (a) isolated nodes decrease with sequence length N; (b) sparsity remains low across all settings.

TABLE XII
EFFICIENCY ANALYSIS OF VARIOUS MODULES IN THE PROPOSED DEFT-DPT FRAMEWORK

Module	Params (M)	GFLOPs	Inference (ms)	GPU Memory (GB)
DEFT + ResNet-50	24.377	125.802	79.707 ± 0.338	0.4493
SVSG	–	–	3.013	–
Co-Attention	8.388	0.0084	~ 0.5	0.0470
GAT	0.528	0.0158	3.271 ± 1.336	0.1253
Full pipeline (RGB + Optical Flow)	**34.218**	**125.827**	$\mathbf{85.991 \pm 2.174}$	**0.5746**

G. Action Localization

A key strength of our DEFT-based framework is its ability to spatially localize actions in egocentric video frames. In Figure 12, DEFT accurately focuses on the relevant hand–object interaction regions for actions such as *"take cereal"*, *"open cupboard"*, and *"move bottle"* on EPIC-Kitchens. Figure 13 shows similar results on Meccano, where *"put_bolt"*, *"take_red_perforated_junction_bar"*, and *"screw_screw_with_screwdriver"* are clearly localized despite background clutter and ego-motion. These examples confirm that DEFT attends to the most discriminative spatial regions, improving both recognition accuracy and interpretability across diverse egocentric datasets.

H. Comparison With Foundation Models

We also examined the efficiency of the proposed framework relative to large-scale pretrained foundation models such as VideoMAE [11], EAKD [12] and EgoVLP [13]. Although these models demonstrate strong generalization,

TABLE XIII

SCALABILITY OF DEFT-DPT AT DIFFERENT INPUT SEQUENCE LENGTHS (N) ON EPIC-KITCHENS DATASET

Frames (N)	GFLOPs	Top-1	F1-score	Inference (ms)	GPU Memory (GB)
30	125.827	0.7049	0.8112	85.99	0.574
100	419.425	0.7053	0.8134	176.84	1.085
150	629.137	0.7087	0.8209	324.09	1.285
200	838.849	0.7094	0.8310	426.86	1.559

they are computationally heavy, containing hundreds of millions of parameters and requiring extensive pretraining. In contrast, our DEFT–DPT network remains lightweight with only 34.218 M parameters (see Table XII), while achieving competitive accuracy on fine-grained egocentric actions. This highlights the framework's ability to effectively capture localized hand–object interactions and temporal dependencies without relying on large-scale multimodal pretraining and compute intensive resources.

I. Execution Time and Scalability

We evaluate the efficiency and scalability of DEFT-DPT on the EPIC-Kitchens dataset. As shown in Table XII, the DEFT+ResNet-50 backbone contains 24.38M parameters and requires 125.80 GFLOPs, processing a 30-frame input in 79.71±0.34 ms. The CPU-based SVSG construction adds only 3.01 ms per window, the Co-Attention module adds 8.388M parameters at negligible cost (0.0084 GFLOPs, ~0.5 ms), and the GAT head adds 0.528M parameters and 0.0158 GFLOPs (3.27±1.34 ms). Overall, DEFT-DPT uses 34.22M parameters and runs in 85.99±2.17 ms per window, reducing computation time by 46.25% compared to SMAST [7].

For scalability, we evaluate sequence lengths from 30 to 200 frames (Table XIII). Although GFLOPs and GPU memory grow with longer sequences due to graph construction, inference time scales reasonably from 85.99 ms ($N = 30$) to 426.86 ms ($N = 200$), while recognition remains stable: Top-1 accuracy ranges from 0.7049 to 0.7094 and the F1-score improves from 0.8112 to 0.8310, showing effective use of longer temporal context. For very long sequences ($N > 500$), we adopt a sliding-window strategy that splits videos into overlapping sub-windows of size W and stride S, processes each independently, and averages the softmax predictions, reducing complexity from $\mathcal{O}(N^2)$ to near-linear $\mathcal{O}(N)$. A preliminary evaluation ($N = 200$, $W = 50$, $S = 25$) achieves a competitive Top-1 accuracy of 0.7094 ± 0.039, demonstrating the practicality of DEFT-DPT for long videos.

J. Graph Connectivity and Sparsity Analysis

We further analyze the proposed Dynamic Percentile Thresholding (DPT) under long sequences and aggressive graph pruning. Specifically, we measure graph sparsity and the percentage of isolated nodes for feasible threshold values from Figure 7, $\tau \in \{94, 95, 96\}$, using 100 random windows from the EPIC-Kitchens (P01_01) dataset, as shown in Table XIV. The results show that the generated graphs remain highly

TABLE XIV

DPT SPARSITY AND CONNECTIVITY ANALYSIS ON EPIC-KITCHENS (P01_01) DATASET

Percentile	Frames (N)	Avg. Edges	Sparsity (%)	Isolated Nodes (%)
94	30	24.0	2.76	45.6
	100	500.1	5.05	7.2
	200	2200.1	5.53	3.4
95	30	16.0	1.84	59.1
	100	400.0	4.04	9.6
	200	1800.0	4.52	4.6
96	30	6.0	0.69	81.9
	100	300.0	3.03	14.1
	200	1400.1	3.52	6.4

sparse, retaining only 0.69%–5.53% of edges, confirming the effectiveness of percentile-based pruning. Moreover, the proportion of isolated nodes decreases with longer sequences due to increased frame diversity. While shorter sequences (N=30) produce more isolated nodes, only 3–6% isolation occur at N=200. In addition, the GAT self-attention mechanism preserves contextual learning even for weakly connected nodes. This is reflected in the stable Top-1 accuracy reported in Table Table XIII, improving slightly from 0.7049 (at N=30) vs. 0.7094 (at N=200).

For qualitative results of action localization and recognition on four publicly available benchmark datasets, please see the supplementary material. These results are also available at https://github.com/pawanesh-mnnit/deft_dpt.

V. CONCLUSION

In this paper, we proposed a lightweight multimodal framework for egocentric action localization and recognition that integrates spatial, temporal, and multimodal cues. The Dynamic Egocentric Feature Transformation (DEFT) adaptively encodes spatially aligned and weighted features, focusing on salient hand–object regions while preserving viewpoint-specific information. The Sparse Video Similarity Graph (SVSG), constructed via Dynamic Percentile Thresholding (DPT), captures the most informative inter-frame relationships while reducing redundancy. A Graph Attention Network (GAT) then enables context-aware learning, yielding compact yet discriminative representations. Experiments on four benchmark egocentric datasets show that our method achieves competitive accuracy while significantly reducing computational cost compared to recent large-scale models. Since the current design assumes that salient hand–object interactions occur near the egocentric viewpoint center, several future directions remain. We plan to extend this framework to action anticipation [22], [23] and continuous temporal segmentation for fine-grained egocentric video understanding, and to make the attention center learnable and guided by object-detection cues for dynamic localization of hand–object interactions in peripheral views. Beyond egocentric video action recognition, the proposed DEFT-DPT framework has the potential to be applied to several related video understanding tasks. The graph-based modeling in the SVSG and GAT modules could be beneficial for group activity recognition, where interactions among multiple individuals need to be captured [42]. The adaptive feature transformation and co-attention fusion mechanisms may also be useful for compositional action recognition, which requires learning relationships between actions and objects [43]. In addition, the proposed co-attention strategy could be extended to other multimodal learning tasks, such as audio-visual event localization [44]. These potential applications highlight the broader applicability of the proposed framework and represent promising directions for future research.

References

[1] H. Wang, J. Yang, B. Yu, Y. Zhan, D. Tao, and H. Ling, "Distilling interaction knowledge for semi-supervised egocentric action recognition," *Pattern Recognit.*, vol. 157, Jan. 2025, Art. no. 110927.

[2] Z. Xing, Q. Dai, H. Hu, J. Chen, Z. Wu, and Y.-G. Jiang, "SVFormer: Semi-supervised video transformer for action recognition," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 18816–18826.

[3] H. Li, W.-S. Zheng, J. Zhang, H. Hu, J. Lu, and J.-H. Lai, "Egocentric action recognition by automatic relation modeling," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 1, pp. 489–507, Jan. 2023.

[4] M. Jaderberg et al., "Spatial transformer networks," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 28, 2015.

[5] M. B. Shaikh, D. Chai, S. M. S. Islam, and N. Akhtar, "From CNNs to transformers in multimodal human action recognition: A survey," *ACM Trans. Multimedia Comput. Commun. Appl.*, vol. 20, no. 8, pp. 1–24, 2024.

[6] A. Núñez-Marcos, G. Azkune, and I. Arganda-Carreras, "Egocentric vision-based action recognition: A survey," *Neurocomputing*, vol. 472, pp. 175–197, Feb. 2022.

[7] M. Korban, P. Youngs, and S. T. Acton, "A semantic and motion-aware spatiotemporal transformer network for action detection," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 9, pp. 6055–6069, Sep. 2024.

[8] H. Wang et al., "Free-form composition networks for egocentric action recognition," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 10, pp. 9967–9978, Oct. 2024.

[9] D. Damen et al., "Rescaling egocentric vision: Collection, pipeline and challenges for EPIC-KITCHENS-100," *Int. J. Comput. Vis.*, vol. 130, no. 1, pp. 33–55, Jan. 2022.

[10] G. Radevski, D. Grujicic, M. Blaschko, M.-F. Moens, and T. Tuytelaars, "Multimodal distillation for egocentric action recognition," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2023, pp. 5190–5201.

[11] Z. Tong, Y. Song, J. Wang, and L. Wang, "VideoMAE: Masked autoencoders are data-efficient learners for self-supervised video pre-training," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 35, 2022, pp. 10078–10093.

[12] N. Zheng, X. Song, T. Su, W. Liu, Y. Yan, and L. Nie, "Egocentric early action prediction via adversarial knowledge distillation," *ACM Trans. Multimedia Comput., Commun., Appl.*, vol. 19, no. 2, pp. 1–21, May 2023.

[13] S. Pramanick et al., "EgoVLPv2: Egocentric video-language pre-training with fusion in the backbone," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2023, pp. 5262–5274.

[14] A. Sahu and A. S. Chowdhury, "Together recognizing, localizing and summarizing actions in egocentric videos," *IEEE Trans. Image Process.*, vol. 30, pp. 4330–4340, 2021.

[15] S. Sudhakaran, S. Escalera, and O. Lanz, "LSTA: Long short-term attention for egocentric action recognition," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 9946–9955.

[16] T. Shiota, M. Takagi, K. Kumagai, H. Seshimo, and Y. Aono, "Egocentric action recognition by capturing hand-object contact and object state," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis. (WACV)*, Jan. 2024, pp. 6527–6537.

[17] X. Zhang, Y. Jia, J. Zhang, Y. Yang, and Z. Tu, "Robust 2D skeleton action recognition via decoupling and distilling 3D latent features," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 35, no. 10, pp. 10410–10422, Oct. 2025.

[18] Y. Li, Z. Ye, and J. M. Rehg, "Delving into egocentric actions," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 287–295.

[19] H. Pirsiavash and D. Ramanan, "Detecting activities of daily living in first-person camera views," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2012, pp. 2847–2854.

[20] Y. Yin, M. Liu, R. Yang, Y. Liu, and Z. Tu, "Dark-DSAR: Lightweight one-step pipeline for action recognition in dark videos," *Neural Netw.*, vol. 179, Nov. 2024, Art. no. 106622.

[21] F. Ragusa, A. Furnari, S. Livatino, and G. M. Farinella, "The MECCANO dataset: Understanding human-object interactions from egocentric videos in an industrial-like domain," in *Proc. IEEE Winter Conf. Appl. Comput. Vis. (WACV)*, Jan. 2021, pp. 1568–1577.

[22] T. Zhang, W. Min, T. Liu, S. Jiang, and Y. Rui, "Toward egocentric compositional action anticipation with adaptive semantic debiasing," *ACM Trans. Multimedia Comput. Commun. Appl.*, vol. 20, no. 5, pp. 1–21, 2024.

[23] C. Plizzari et al., "An outlook into the future of egocentric vision," *Int. J. Comput. Vis.*, vol. 132, no. 11, pp. 4880–4936, 2024.

[24] M. Hatano, R. Hachiuma, R. Fujii, and H. Saito, "Multimodal cross-domain few-shot learning for egocentric action recognition," in *Proc. Eur. Conf. Comput. Vis.*, 2024, pp. 182–199.

[25] L. Xu et al., "Towards continual egocentric activity recognition: A multi-modal egocentric activity dataset for continual learning," *IEEE Trans. Multimedia*, vol. 26, pp. 2430–2443, 2023.

[26] B. Zhou, P. Wang, J. Wan, Y. Liang, and F. Wang, "A unified multimodal de{-} and re-coupling framework for RGB-D motion recognition," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 10, pp. 11428–11442, Oct. 2023.

[27] W. Wang, Y. Su, and J. Gu, "TransCLIP: Transferring vision–language models for efficient video action recognition," *IEEE Trans. Ind. Informat.*, vol. 21, no. 10, pp. 7643–7652, Oct. 2025.

[28] H. Xu, Y. Gao, Z. Hui, J. Li, and X. Gao, "Language knowledge-assisted representation learning for skeleton-based action recognition," *IEEE Trans. Multimedia*, vol. 27, pp. 5784–5799, 2025.

[29] J. Zhang, Z. Tu, J. Weng, J. Yuan, and B. Du, "A modular neural motion retargeting system decoupling skeleton and shape perception," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 10, pp. 6889–6904, Oct. 2024.

[30] Y. Zou, C. D. Nugent, M. Burns, S. Wu, L. Zhu, and M. Liu, "Adaptive attention-based unsupervised domain adaptation for egocentric action recognition," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 36, no. 5, pp. 5852–5864, May 2026.

[31] T. Liu, K.-M. Lam, R. Zhao, and J. Kong, "Enhanced attention tracking with multi-branch network for egocentric activity recognition," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 32, no. 6, pp. 3587–3602, Jun. 2022.

[32] W. Hamilton, Z. Ying, and J. Leskovec, "Inductive representation learning on large graphs," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2017.

[33] X. Yue, G. Qu, and L. Gan, "GIN-graph: A generative interpretation network for model-level explanation of graph neural networks," 2025, *arXiv:2503.06352*.

[34] P. Huang, R. Yan, X. Shu, Z. Tu, G. Dai, and J. Tang, "Semantic-disentangled transformer with noun-verb embedding for compositional action recognition," *IEEE Trans. Image Process.*, vol. 33, pp. 297–309, 2024.

[35] G. Dai, X. Shu, W. Wu, R. Yan, and J. Zhang, "GPT4Ego: Unleashing the potential of pre-trained models for zero-shot egocentric action recognition," *IEEE Trans. Multimedia*, vol. 27, pp. 401–413, 2025.

[36] J. Chalk, J. Huh, E. Kazakos, A. Zisserman, and D. Damen, "TIM: A time interval machine for audio-visual action recognition," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2024, pp. 18153–18163.

[37] J. Kini, S. Fleischer, I. Dave, and M. Shah, "Egocentric RGB+Depth action recognition in industry-like settings," 2023, *arXiv:2309.13962*.

[38] M. Lu, Z.-N. Li, Y. Wang, and G. Pan, "Deep attention network for egocentric action recognition," *IEEE Trans. Image Process.*, vol. 28, no. 8, pp. 3703–3713, Aug. 2019.

[39] S. Singh, C. Arora, and C. V. Jawahar, "First person action recognition using deep learned descriptors," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 2620–2628.

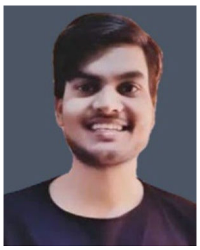
[40] Y. Shen, B. Ni, Z. Li, and N. Zhuang, "Egocentric activity prediction via event modulated attention," in *Proc. Eur. Conf. Comput. Vis.*, 2018, pp. 197–212.

[41] G. Kapidis, R. Poppe, and R. C. Veltkamp, "Multi-dataset, multitask learning of egocentric vision tasks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 6, pp. 6618–6630, Jun. 2023.

[42] R. Yan, L. Xie, J. Tang, X. Shu, and Q. Tian, "HiGCIN: Hierarchical graph-based cross inference network for group activity recognition," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 6, pp. 6955–6968, Jun. 2023.

[43] R. Yan, L. Xie, X. Shu, L. Zhang, and J. Tang, "Progressive instance-aware feature learning for compositional action recognition," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 8, pp. 10317–10330, Aug. 2023.

[44] L. Xing, H. Qu, R. Yan, X. Shu, and J. Tang, "Locality-aware cross-modal correspondence learning for dense audio-visual events detection," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 36, no. 5, pp. 6838–6851, May 2026.



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